Implementation Plan: N8N Hands-On Intermediate Assessment

Topic 14: Sub-Workflows, Debugging & Production Readiness

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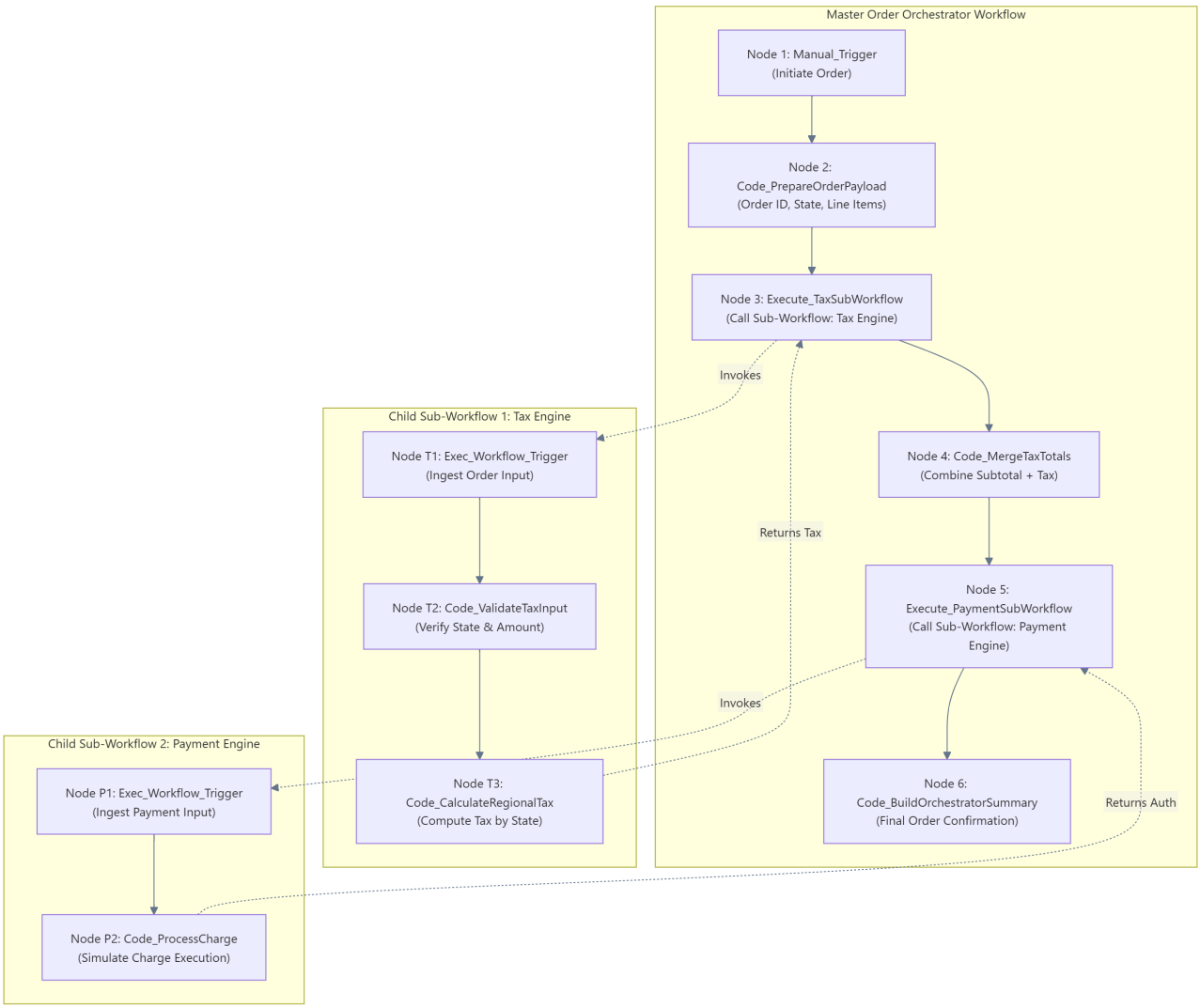
Concept Guide | Requirement Document | **Step-by-Step Implementation Plan** | Test Plan

1. Objective

Build a modular master order orchestration workflow in n8n that delegates domain logic to specialized sub-workflows (Tax Calculation Engine, Payment Gateway Processor) using the Execute Workflow node. Implement strict input contract validation, demonstrate execution tracing and node pinning for debugging, and provide a production-ready Docker Compose environment blueprint.

2. Canvas Architecture Overview

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3. Prerequisites & Environment Setup

- [ ] n8n instance version 1.0.0 or higher.
- [ ] Production-hardened `docker-compose.yml` snippet (provided below):

```
version: '3.8'
services:
  postgres:
    image: postgres:15-alpine
    environment:
      POSTGRES_DB: n8n_prod
      POSTGRES_USER: n8n_admin
      POSTGRES_PASSWORD: StrongProductionPassword2026!
    volumes:
```

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```
      - postgres_storage:/var/lib/postgresql/data
    restart: always

  n8n:
    image: n8nio/n8n:latest
    ports:
      - "5678:5678"
    environment:
      - DB_TYPE=postgresdb
      - DB_POSTGRESDB_HOST=postgres
      - DB_POSTGRESDB_PORT=5432
      - DB_POSTGRESDB_DATABASE=n8n_prod
      - DB_POSTGRESDB_USER=n8n_admin
      - DB_POSTGRESDB_PASSWORD=StrongProductionPassword2026!
      - N8N_ENCRYPTION_KEY=e83a9d20c5b41f87a6d102e93847ab5c1209ef48b9a1
      - EXECUTIONS_DATA_PRUNE=true
      - EXECUTIONS_DATA_MAX_AGE=168
      - EXECUTIONS_DATA_PRUNE_TIMEOUT=3600
      - GENERIC_TIMEZONE=UTC
    volumes:
      - n8n_data:/home/node/.n8n
    depends_on:
      - postgres
    restart: always

volumes:
  postgres_storage:
  n8n_data:
```

---

4. Step-by-Step Practical Build Guide

Step 1: Build Child Sub-Workflow 1 - Tax Calculation Engine

1. **Create Sub-Workflow:**
   - In your n8n dashboard, click **+ Add workflow**.
   - Rename in top-left to: `Sub-Workflow 1 - Tax Engine`.
2. **Add Execute Workflow Trigger:**
   - Click **+ Add first step...**, search for **`Execute Workflow Trigger`**, and place it.
   - Rename to: `Execute_Workflow_Trigger`.
3. **Add State Tax Calculation Engine:**
   - From `Execute_Workflow_Trigger`, click **`+`**, search for **`Code`**, and add the node.
   - Rename to: `Code_CalculateStateTax`.
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste:

```
const order = $input.item.json;
const stateRates = { 'CA': 0.0825, 'NY': 0.08875, 'TX': 0.0625, 'FL': 0.0600 };
```

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```
const rate = stateRates[order.state] || 0.0500; // 5% default

const subtotal = Number(order.subtotal || 0);
const taxAmount = Number((subtotal * rate).toFixed(2));
const grandTotal = Number((subtotal + taxAmount).toFixed(2));

return {
  json: {
    orderId: order.orderId,
    subtotal: subtotal,
    state: order.state,
    appliedTaxRate: rate,
    taxAmount: taxAmount,
    grandTotal: grandTotal,
    taxCalculatedAt: new Date().toISOString()
  }
};
```

4. **Save & Copy Workflow ID:**
   - Click **Save** in the top-right toolbar.
   - Note the URL in your browser bar (e.g. `http://.../workflow/<subworkflow-id-1>`). Copy this ID.

Step 2: Build Child Sub-Workflow 2 - Payment Gateway Processor

1. **Create Sub-Workflow:**
   - Click **+ Add workflow**. Rename to: `Sub-Workflow 2 - Payment Gateway`.
2. **Add Trigger & Payment Execution:**
   - Add **Execute Workflow Trigger** node named `Execute_Workflow_Trigger`.
   - From trigger, add **HTTP Request** node named `HTTP_ProcessPaymentCharge`.
     - **Method:** Select **`POST`** (Mode: **Fixed**).
     - **URL:** Toggle to **Expression (`fx`)**, enter: `https://dummyjson.com/c/<your-id>` (or mock payment endpoint).
     - **Send Body:** Switch toggle to **`ON`**.
     - **Body Content Type:** Select **`JSON`**.
     - **Specify Body:** Select **`Using JSON`**.
     - **JSON:** Switch to **Expression (`fx`)** and enter:

```
{{ { "orderId": $json.orderId, "amount": $json.grandTotal } }}
```

3. **Add Response Formatter:**
   - Connect to a **Code** node named `Code_FormatPaymentReceipt`:

```
const resp = $input.item.json;
```

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```
return {
  json: {
    paymentStatus: "CONFIRMED",
    transactionRef: "AUTH-" + Math.random().toString(36).substring(2, 9).toUpperCase(),
    amountCharged: $node['Execute_Workflow_Trigger'].json.grandTotal,
    processedAt: new Date().toISOString()
  }
};
```

4. **Save & Copy Workflow ID:**
   - Click **Save** and note `<subworkflow-id-2>`.

Step 3: Build the Master Order Orchestrator Workflow

1. **Create Master Workflow & Trigger:**
   - Click **+ Add workflow**, rename to: `Topic 14 - Sub-Workflows, Debugging & Production Readiness`.
   - Add a **Manual Trigger** node named `Manual_Trigger`.
2. **Add Master Order Stager Node:**
   - From `Manual_Trigger`, click `+`, search for **`Code`**, and add the node.
   - Rename to: `Code_StageMasterOrder`.
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for All Items`**.
   - **JavaScript Code:** Paste:

```
return [
  {
    json: {
      orderId: "ORD-MODULAR-901",
      customerName: "Acme Corporation",
      state: "CA",
      subtotal: 1000.00
    }
  }
];
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify staged order appears in Output.

Step 4: Invoke Tax Sub-Workflow (**`Execute_TaxSubWorkflow`**)

1. **Add Node to Canvas:**
   - From `Code_StageMasterOrder`, click `+`, search for **`Execute Workflow`**, and select the **Execute Workflow** node.

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- Rename to: `Execute_TaxSubWorkflow`.

2. **Configure Parameters (Parameters Tab):**
   - **Source:** Select **`Database`** (or `From list`).
   - **Workflow:** Select `Sub-Workflow 1 - Tax Engine` (or enter `<subworkflow-id-1>`).
   - **Mode:** Select **`Once for each item`**.
3. **Test Node Checkpoint:**
   - Click **Test step**. Notice that the Tax Sub-Workflow runs remotely and returns `appliedTaxRate: 0.0825`, `taxAmount: 82.50`, and `grandTotal: 1082.50`!

Step 5: Merge Tax Outputs (`Code_MergeTaxTotals`)

1. **Add Node to Canvas:**
   - From `Execute_TaxSubWorkflow`, click **`+`**, search for **`Code`**, and add the node.
   - Rename to: `Code_MergeTaxTotals`.
2. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste:

```
const order = $node['Code_StageMasterOrder'].json;
const taxResult = $input.item.json;

return {
  json: {
    ...order,
    taxRate: taxResult.appliedTaxRate,
    taxAmount: taxResult.taxAmount,
    grandTotal: taxResult.grandTotal
  }
};
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify merged order contains `grandTotal: 1082.5`.

Step 6: Invoke Payment Sub-Workflow (`Execute_PaymentSubWorkflow`)

1. **Add Node to Canvas:**
   - From `Code_MergeTaxTotals`, click **`+`**, search for **`Execute Workflow`**, and add the node.
   - Rename to: `Execute_PaymentSubWorkflow`.
2. **Configure Parameters (Parameters Tab):**
   - **Source:** Select **`Database`**.

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- **Workflow:** Select `Sub-Workflow 2 - Payment Gateway` (or enter `<subworkflow-id-2>`).
- **Mode:** Select **`Once for each item`**.

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify the payment sub-workflow executes and returns `paymentStatus: "CONFIRMED"`.

Step 7: Build Final Orchestrator Summary (`Code_BuildOrchestratorSummary`)

1. **Add Node to Canvas:**
   - From `Execute_PaymentSubWorkflow`, click **`+`**, search for **`Code`**, and add the node.
   - Rename to: `Code_BuildOrchestratorSummary`.
2. **Configure Parameters (Parameters Tab):**
   - **Language:** Select **`JavaScript`**.
   - **Mode:** Select **`Run Once for Each Item`**.
   - **JavaScript Code:** Paste:

```
const payment = $input.item.json;
const order = $node['Code_MergeTaxTotals'].json;

return {
  json: {
    orchestrationResult: "ORDER_FULFILLED_VIA_MICROSERVICES",
    orderId: order.orderId,
    customer: order.customerName,
    grandTotalPaid: order.grandTotal,
    authCode: payment.transactionRef,
    executedSubWorkflows: [
      "Sub-Workflow 1: Tax Engine",
      "Sub-Workflow 2: Payment Gateway"
    ],
    completedAt: new Date().toISOString()
  }
};
```

3. **Test Node Checkpoint:**
   - Click **Test step**. Verify consolidated master order execution receipt.

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5. Node-by-Node Settings Blueprint

Node ID	Node Name	Node Type	Critical Settings	Custom Expressions / Code
N14-01	`Manual_Trigger`	Manual Trigger	Trigger execution on demand	N/A

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N14-02	`Code_PrepareOrderPayload`	Code	Emit order with state NY and subtotal 1200	Order fixture payload
N14-03	`Execute_TaxSubWorkflow`	Execute Workflow	Target: Tax Sub-Workflow	Passes `{ state, subtotal }`
N14-04	`Code_MergeTaxTotals`	Code	Combines order fields with sub-workflow tax return	Merges sub-workflow output
N14-05	`Execute_PaymentSubWorkflow`	Execute Workflow	Target: Payment Sub-Workflow	Passes `{ orderId, amount: grandTotal }`
N14-06	`Code_BuildOrchestratorSummary`	Code	Master orchestration confirmation record	Final audit formatter

6. Verification Checkpoints & Testing Guide

Test 1: Full Modular Orchestrator Execution

1. Ensure both child sub-workflows are saved in your n8n workspace.
2. In `Topic 14 - Sub-Workflows & Modular Orchestrator`, click **Test Workflow**.
3. Inspect `Execute_TaxSubWorkflow` output:
   - State `NY` (Rate: `0.08875`) on Subtotal `$1,200.00` yields `taxAmount: 106.50`, `grandTotal: 1306.50`.
4. Inspect `Execute_PaymentSubWorkflow` output:
   - Successfully processes charge for `$1,306.50`, returning valid `chargeId: ch_sub_...`.
5. Verify `Code_BuildOrchestratorSummary` outputs `orchestratorStatus: "ORDER_ORCHESTRATION_COMPLETED"` with exact financial summary.

Test 2: Sub-Workflow Contract Enforcement (Isolation Test)

1. Open `Sub-Workflow: Tax Calculation Engine` in its own canvas.
2. Click **Test Step** with an empty payload `{}`.
3. Verify `Code_ValidateTaxInput` throws `CONTRACT_VIOLATION: Missing required parameter 'state'`.
4. Demonstrates that sub-workflows protect themselves against corrupt or incomplete caller input.

7. Production Risks & Mitigation Strategies

Risk	Impact	Mitigation
Deleted Sub-Workflow ID	Orchestrator crashes if a child workflow is deleted or recreated with a new ID.	Call sub-workflows by name or maintain ID lookup constants in global static data.
Sub-Workflow Execution Timeout	Long-running child process holds parent thread, timing out caller.	For non-blocking operations, trigger sub-workflows asynchronously via Webhooks or queues.
Circular Invocation Deadlock	Workflow A calls Workflow B which calls Workflow A, exhausting stack memory.	Enforce strict hierarchical orchestration: parent flows call children; child flows never call parents.

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